

Heaven's Light is Our Guide

RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY

RAJSHAHI-6204, BANGLADESH



Industrial Attachment Report

Group – 06

SQ Hues LTD.

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ABSTRACT

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BACKGROUND OF THE INDUSTRY

SQ Group, established in 1993, has grown into one of the leading apparel manufacturing organizations in Bangladesh. The company operates under a fully integrated business model, enabling control over every stage of the production process from raw material processing to the delivery of finished garments. This integrated approach ensures consistent quality, operational efficiency, and timely delivery to global markets.

Manufacturing Units and Vertical Integration

The manufacturing strength of SQ Group is built upon three specialized business units, each designed to serve distinct product categories and operational requirements.

SQ Celsius Limited is dedicated to the production of high-quality knitwear, with a primary focus on sweaters. The unit emphasizes precision manufacturing and superior finishing standards.

SQ Birichina Limited specializes in the production of premium lingerie, maintaining high standards of comfort, design, and quality to meet international expectations.

SQ Hues Limited functions as the Group's wet processing and vertical support facility. It provides essential back-end services, including yarn dyeing, garment washing, fabric knitting and dyeing, and garment dyeing.

This vertical integration enables SQ Group to maintain strict quality control, reduce dependency on external suppliers, and significantly shorten production lead times, ensuring efficiency across the entire supply chain.

Product Portfolio and Global Reach

SQ Group primarily focuses on the production of sweaters and lingerie, catering to the demands of high-end international markets. The organization is committed to maintaining superior product quality, aligning with global standards and customer expectations.

The Group has established a strong presence in international markets, exporting its products to various regions around the world. Key export destinations include countries in Europe such as the United Kingdom, Sweden, Denmark, and Spain; North America, particularly the United States; Oceania, including Australia; and Asia, with Japan as a major market. This wide geographic reach reflects the company's reliability and competitiveness in the global apparel industry.

Conclusion

Since its establishment in 1993, SQ Group has demonstrated consistent growth through its commitment to quality, innovation, and vertical integration. By operating specialized

manufacturing units supported by the advanced processing capabilities of SQ Hues Limited, the Group continues to strengthen its position as a trusted partner for leading global fashion brands. Its ability to deliver end-to-end manufacturing solutions ensures sustained competitiveness in the international apparel market.

CHAPTER 1

Introduction

1.1 Introduction

SQ Hues Limited is a modern textile wet processing facility and a key subsidiary of the SQ Group, one of Bangladesh's leading vertically integrated apparel manufacturers. Located in Valuka, the company plays a vital role in the group's supply chain by providing high-quality dyed and finished textile products. As the wet processing unit of the group, SQ Hues specializes in fabric dyeing, yarn dyeing, garment dyeing, and advanced washing techniques. It supports sister concerns such as SQ Birichina and SQ Celsius, ensuring smooth backward linkage and operational efficiency within the integrated production system. The facility is equipped with advanced machinery and modern dyeing technologies, enabling it to handle a wide range of fibers including cotton, blends, and specialty fabrics while maintaining international quality standards. SQ Hues also emphasizes sustainable production practices, focusing on efficient resource utilization and environmentally responsible processing methods. Overall, SQ Hues Limited serves as a critical backbone of SQ Group's vertical integration, contributing to improved product quality, faster production lead time, and enhanced competitiveness in the global textile and apparel market.

The Main operations conducted in SQ hues are

- Knitting
- Fabric Dying
- Soft winding
- Hard Winding
- Yarn Dying
- Finishing
- Quality Check
- Washing
- Color Development & R & D
- Physical Testing Lab
- Planning

CHAPTER 2

Circular Knitting Section

2.1 Introduction

Circular knitting is a fabric manufacturing process in which yarn is continuously interloped to form a seamless tubular structure. Unlike flat knitting, where fabric is produced in rows and requires side seams, circular knitting creates fabric in a spiral or round form using needles arranged in a circular configuration. The process can be carried out either manually using circular needles or industrially using circular knitting machines.

In industrial applications, circular knitting machines operate with multiple needles mounted on a rotating cylinder. As the yarn is fed into the machine, the needles move in a synchronized motion to form loops, resulting in a continuous tube of fabric. This method allows high production speed, uniform fabric quality, and minimal material wastage.

Circular knitting is widely used in the textile industry for producing garments such as T-shirts, socks, hosiery, and sportswear due to its seamless construction, elasticity, and comfort. The absence of seams enhances the strength and aesthetic quality of the fabric, making it highly suitable for mass production and modern apparel manufacturing.

2.2 Plant Layout of Knitting Section

The knitting section is systematically organized into five columns of machinery, covering units numbered 1 to 27, which are arranged to ensure smooth production flow and efficient space utilization. These columns include both primary production machines and specialized machinery such as interlock and Blix systems. In addition to the production area, the section also includes a dedicated store for materials and spare parts, a quality inspection area for monitoring fabric standards, and an administrative office for operational control and coordination. Furthermore, a griage inspection area is provided to examine raw knitted fabric before further processing, ensuring consistency, defect detection, and overall quality assurance.

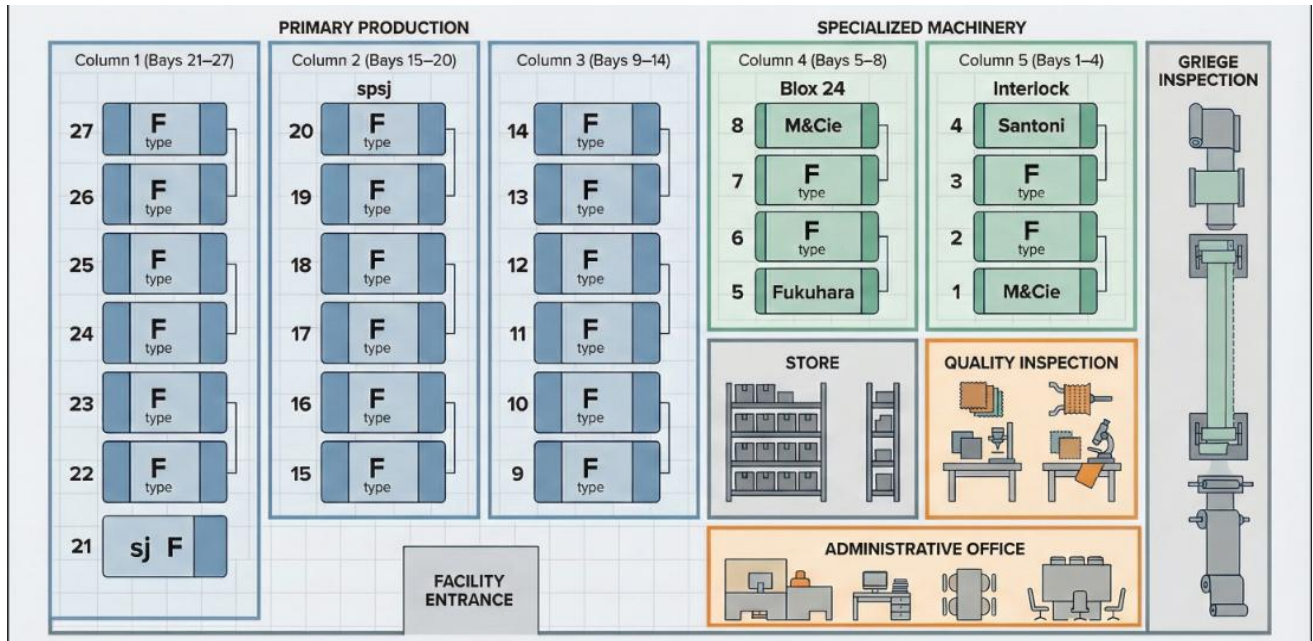


Figure 1 Plant Layout of Sq hues Knitting section

2.3 Knitting Production Cycle

Knitting is a method of fabric construction created by interlacing loops of yarn together using needles.

Knitting Machine Brands and Types

There are three types of knitting Machine used in Sq hues and they are:

- Mayer & Cie
- Fukuhara
- Santoni



Figure 2 Knitting Machine Used in SQ Hues

These machines are categorized by the type of fabric they produce, specifically:

- **Single Jersey machines:** They use a single needle bed (the cylinder) and sinkers to hold the loops down while the needles rise to pick up new yarn.
- **Double Jersey machines:** It features two needle beds, a vertical cylinder, and a horizontal Dial. The needles from the dial and cylinder intermesh, allowing the machine to knit two layers of fabric simultaneously that are joined together.

Flow Chart of Knitting:

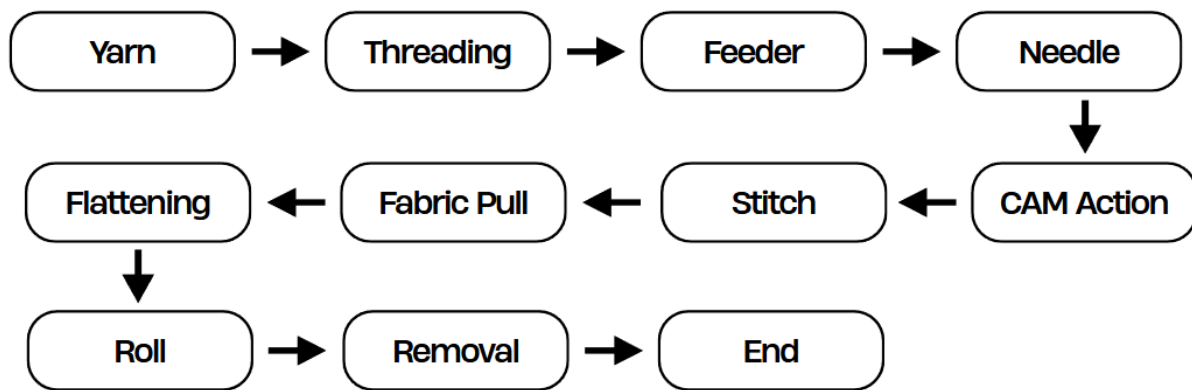


Figure 3 Knitting Flow Chart

2.4 Knitting Production Formula

$$\text{Production} = \frac{\text{Number of Needles} \times \text{RPM} \times \text{Efficiency} \times 8 \times \text{hrs} \times 60 \times 1.02}{\text{Count} \times 840 \times 1000 \times 2.204}$$

Here.

Needle: The number of needles represents the total needles present in the circular knitting machine. A higher number of needles increases the number of loops formed simultaneously, which directly increases the fabric production rate.

RPM: RPM (revolutions per minute) indicates the rotational speed of the machine cylinder. As RPM increases, the machine produces more loops per minute, thereby increasing overall production.

Efficiency: Efficiency refers to the actual operating performance of the machine. It accounts for production losses due to machine stoppages, maintenance, or operational delays, and is usually expressed as a percentage.

8x60: The terms 8 and 60 represent the total working time. Here, 8 indicates the number of working hours per shift, and multiplying by 60 converts the time into minutes for calculation purposes.

1.02: The factor 1.02 is an allowance factor added to compensate for minor losses such as yarn wastage, machine variations, and other small inefficiencies during production.

Count: The count refers to the yarn fineness in the English cotton count system. A higher count indicates finer yarn, which results in lower weight per unit length of fabric.

840: The constant 840 is a standard value used in the yarn count system, representing the number of yards per pound in the English count system.

The factor 1000 is used as a unit conversion factor, typically to convert the calculated value into kilograms or to scale the production quantity.

The value 2.204 is a conversion factor used to convert weight from kilograms to pounds, ensuring consistency in unit calculations.

2.5 Knitting Defects

The Defects are categorized in different segments such as-

Machine & Mechanical Faults

- **Knitting Hole:** Structural gaps in the fabric are often caused by needle breakage or yarn tension issues.
- **Lycra Drop:** A specific defect occurring in stretch fabrics when the elastic component is not properly integrated.
- **Latch Mark:** Marks left on the fabric caused by a malfunctioning latch on a knitting needle.
- **Barry Mark (Barre):** Horizontal streaks or bands in the fabric caused by variations in yarn or machine settings.

Yarn-Related Defects

- **Thick Cotton:** Areas where the yarn used is noticeably thicker than the rest of the fabric.
- **Slub:** A lump or thick place in the yarn that appears as a defect in the finished knit.

- **Dead Cotton:** Fibers that do not absorb dye properly (often appearing as light spots later).
- **Lot Mix:** A defect caused by mixing different batches or lots of yarn, leading to inconsistencies in appearance or dye uptake.

Contamination & Stains

- **Yarn Contamination:** Foreign fibers or materials that have been spun into the yarn.
- **Oil Spot & Oil Mark:** Stains caused by lubricants from the knitting machinery leaking onto the fabric



Figure 4 Knitting Faults

2.6 Quality Control: Greige Fabric Inspection

All fabric undergoes a Greige **Inspection** to identify and rectify defects before the material proceeds to dyeing. The facility utilizes specialized **inspection machines equipped with high-intensity backlit screens**. Operators monitor the moving fabric for common knitting issues, including:

- **Holes or drop stitches.**
- **Needle lines and yarn irregularities.**
- **Oil stains or contamination.**

CHAPTER 3

Dyeing Section

3.1 Introduction

The dyeing section is a crucial part of the textile manufacturing process where raw or greige fabric is treated to achieve the desired color, appearance, and finish. In this stage, fabrics produced from the knitting section are processed using various dyeing techniques and chemicals to ensure uniform coloration and high-quality output. The section is equipped with specialized dyeing machines, chemical dosing systems, and temperature control units to maintain precise processing conditions. Proper control of parameters such as temperature, time, pH, and chemical concentration is essential to achieve consistent shade and color fastness. In addition to coloration, the dyeing process also enhances fabric properties such as softness, durability, and resistance to washing and light. The dyeing section plays a vital role in determining the final aesthetic and functional quality of the textile product, making it an integral component of the overall production system.

In sq Hues, usually two types of dying occur, such as

- Fabric Dying
- Yarn Dying

3.2 Fabric Dying

The **fabric dyeing process** is a multi-stage chemical and mechanical procedure designed to impart uniform color and remove impurities from textiles.



Figure 5 Fabric Dying Section

3.3 Dyeing Flow Chart:

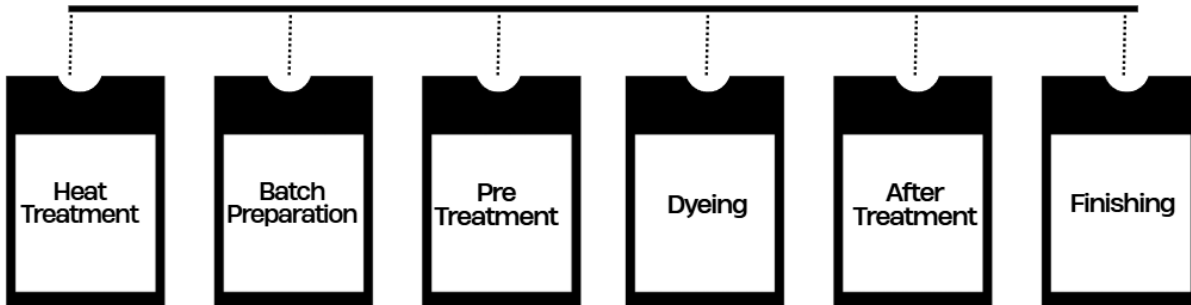


Figure 6 Dyeing Flow Chart

Heat Setting:

Heat setting is a thermal process used in textile manufacturing to stabilize the structure and dimensions of synthetic and blended fabrics. In this process, the fabric is exposed to a controlled high temperature for a specific period, allowing the fibers to relax and set in their desired form. This treatment is especially important for materials such as polyester, nylon, and their blends, which are sensitive to heat and can retain a new shape after thermal exposure.

The main purpose of heat setting is to improve dimensional stability, reduce shrinkage, and prevent unwanted deformation during subsequent processes like dyeing, washing, and finishing. It also helps in enhancing fabric properties such as wrinkle resistance, smoothness, and shape retention. Proper control of temperature, tension, and processing time is essential to achieve uniform results and avoid fabric damage. Overall, heat setting ensures better quality, durability, and performance of the final textile product.

Batch Preparation:

Batch preparation ensures that the fabric is correctly identified and organized according to the capacity of the dyeing equipment. The process follows these four steps:

1. **Receive Fabric from Grige Store:** The raw, unbleached fabric (referred to as "greige" or greige) is collected from the storage facility.
2. **Check Sticker & Weight:** Technicians verify the identifying stickers, and the physical weight of the fabric rolls to ensure they match the production order. This aligns with the "batch check" mentioned in the previous dyeing flowchart.

3. **Prepare Batch according to M/C Nozzle:** The fabric is organized into batches based on the number and capacity of the Machine (M/C) Nozzles. This step is vital because the volume of fabric must match the machine's mechanical capabilities to ensure an even distribution of dye.
4. **Arrive at Dyeing Machine:** Once the batch is weighed and organized, it is transported to the dyeing machine station

Pre Treatment:

The process begins with rigorous quality control and cleaning to ensure the fabric is ready to absorb dye evenly.

- **Verification:** A manual **batch, color, and number check** is conducted before loading to ensure consistency.
- **Loading:** The fabric is introduced into the dyeing machine using a **1:6 liquoration ratio**, which defines the weight of the fabric relative to the volume of the dye bath.
- **Scouring:** To remove natural oils and impurities, the fabric is scoured at high temperatures, either at **98°C for 60 minutes** or **110°C for 30 minutes**.
- **Initial Neutralization:** The fabric is **neutralized at 50°C for 15 minutes** and then **washed** to stabilize the fibers.

The Dyeing Phase

This stage involves the actual application of color and chemical preparation.

- **Demineralization:** The fabric undergoes demineralization to remove any mineral content that might interfere with the dye.
- **Color Application:** The **dyeing** itself is performed by mixing the fabric with "**soda color**" to achieve the target shade.

Post-Dyeing and Finishing

Final steps are taken to ensure colorfastness and prepare the fabric for final use.

- **Neutralizing and Soaping:** The fabric is **neutralized** a second time and then undergoes **soaping** to wash away any unfixed dye particles.

- **Mechanical Finishing:** The process mentions **slitting** and the use of a **stenter**, which are standard industrial methods for opening the fabric and heat-setting it to its final width and texture.

3.4 Dyeing Machines

There are two types of dyeing machines used in SQ, which play a vital role in ensuring efficient and high-quality fabric dyeing operations. These machines are selected based on their performance, reliability, and ability to handle different types of fabrics and dyeing requirements.

1. The first type is the **Fong dyeing** machine, with a total of 8 units installed in the dyeing section. These machines are widely known for their advanced technology, energy efficiency, and consistent dyeing performance. Fong machines provide precise control over temperature, pressure, and chemical dosing, which helps achieve uniform coloration and better fabric quality.



Figure 7 Fong's Dyeing Machine

2. The second type is the **Thies dyeing** machine, with 12 units available in the facility. Thies machines are recognized for their durability, high productivity, and flexibility in processing various fabric types. They are designed to ensure excellent dye penetration, reduced water consumption, and improved overall efficiency.



Figure 8 Thies Dyeing Machine

Together, these machines enhance the production capacity of the dyeing section while maintaining high standards of quality, consistency, and operational efficiency.

3.5 Defects of dyeing

There are usually three types of dyeing defects in fabric: Dying. They are-

- **Shade Variation:** This refers to inconsistencies in the color of the fabric, where different areas or batches do not match the intended target color or each other.
- **Speckiness:** This defect involves the appearance of small, unintended spots or "specks" of concentrated color or impurities on the fabric surface.
- **Crocking / Poor Colorfastness:** Crocking occurs when the dye transfers from the fabric surface onto other surfaces through rubbing. This is directly linked to poor colorfastness, meaning the fabric does not retain its color well during use or cleaning.

3.6 Dyeing Production Capacity

Formula

$$\text{Production Capacity} = \frac{\text{Batch Capacity (kg)} \times \text{Efficiency} \times 24}{\text{Total Cycle Time (hr)}}$$

Here,

- **Batch Capacity (kg)** means How much fabric the machine holds in one batch

- **Efficiency** means Machine working efficiency (example: 0.85 or 85%)
- **24** means Hours in a day
- **Total Cycle Time** means Time needed for one dyeing cycle

Total Water (Liquor) Calculation

- Total Water = Fabric Weight (kg) × Liquor Ratio

Chemical Amount Calculation

- Chemical (g) = Fabric Weight (kg) × Concentration (g/L) × Liquor Ratio

Required Dye Calculation

Formula

$$\text{Required Dye (g)} = \frac{\text{Shade \%} \times \text{Fabric Weight (g)}}{100}$$

CHAPTER 4

Finishing Section

4.1 Introduction

The finishing stage begins after the dyeing process and focuses on achieving the desired physical characteristics and quality standards of the fabric. In this stage, the dyed fabric undergoes a series of mechanical and chemical treatments to enhance its appearance, texture, and performance. Processes such as drying, heat setting, compacting, and calendaring are commonly applied to improve dimensional stability, smoothness, and overall finish.

Finishing also plays a crucial role in controlling fabric properties such as shrinkage, softness, strength, and wrinkle resistance. Special chemical finishes may be applied to provide additional functionalities like water repellency, anti-pilling, or improved durability. Careful control of process parameters such as temperature, pressure, and machine speed is essential to ensure consistent quality. Overall, the finishing stage ensures that the fabric meets customer requirements and is ready for final use or garment production.

4.2 Finishing Flow Chart

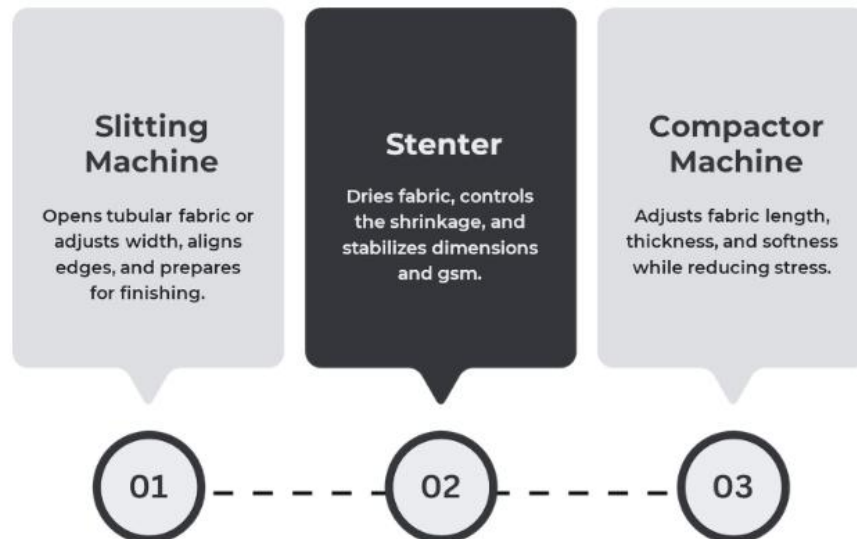


Figure 9 Finishing Flow Chart

4.3 Hydro Extraction

Process: This step involves removing excess water from the fabric, often through squeezing.

Objective: The primary goal is to reduce the energy required for drying in subsequent stages by mechanically extracting as much moisture as possible.

4.4 Slitting

Process: The fabric moves through a slitting machine where tubular fabric is opened. The detailed workflow includes feeding, squeezing, edge guiding, tubular slitting, opening width, and selvage alignment before batching.

Objective: This step prepares the fabric for finishing by adjusting its width and ensuring the edges are properly aligned



Figure 10 Slitting Machine

Slitting Flow Process Description

The slitting machine is used to open tubular fabric, adjust its width, and align its edges to prepare it for further finishing. Based on the detailed technical layout, the flow occurs as follows:

1. Feeding: The dyed, wet fabric is fed into the machine.
2. Squeezing: The fabric passes through rollers to perform initial hydro extraction.
3. Edge Guiding: The machine ensures the fabric is centred and properly oriented before cutting.
4. Tubular Slitting: This is the central step where the tubular fabric is cut open.

5. Opening Width: The cut fabric is spread out to its full open-width state.
6. Selvage Alignment: The edges (selvedges) of the opened fabric are aligned to ensure uniformity.
7. Batching: The opened and aligned fabric is collected (batched) to be moved to the Stenter machine (e.g., a Brückner) for drying and heat setting.

4.5 Stentering

Process: Often performed using a Brückner machine, this stage involves several sub-steps: pinning/clipping the fabric, setting overfeed, heat drying, and heat setting, followed by a cooling zone and bow/skew correction.

Objective: The stenter dries the fabric while controlling shrinkage. It is critical for stabilizing dimensions and GSM.



Figure 11 Stentering Machine

Stentering Flow Process

The detailed workflow of the stentering process includes the following steps:

Fabric Feeding: The fabric is introduced into the machine after the initial preparation stages.

Padding (Optional): This step is used specifically for special fabrics to apply necessary treatments.

Pinning/Clipping: The fabric edges are secured using pins or clips to maintain the required width and tension during processing.

Overfeed Setting: This setting is adjusted to control the amount of fabric fed into the machine, which is essential for managing shrinkage.

Heat Drying & Heat Setting: The fabric undergoes thermal treatment to remove moisture and permanently "set" its physical dimensions.

Cooling Zone: After being subjected to high heat, the fabric passes through a cooling zone to stabilize it before exiting the machine.

Bow/Skew Correction: The machine corrects any distortions in the fabric grain to ensure it is straight and uniform.

Batching: Finally, the processed fabric is collected in batches, ready to move to the compactor machine (such as a Lafer) for further finishing.

During this stage, precise adjustments can be made to meet customer specifications, such as reducing the fabric width from an initial 71 inches to a target of 68/69 inches

4.6 Compaction

Compaction is a critical mechanical finishing stage that typically follows stentering in the fabric production flow.

Equipment Used: This stage utilizes a compactor machine, with the Lafer brand specifically identified as the equipment used for this purpose.

Primary Objectives: The machine is designed to adjust the fabric's length, thickness, and softness. Furthermore, it performs the vital function of reducing internal stress within the fabric to ensure its final stability and quality.



Figure 12 Compaction Machine

Position in the Workflow:

- **Preceding Step:** It occurs after the fabric has passed through the Stenter machine (e.g., Brückner), where the material was dried and its dimensions and GSM were stabilized.
- **Subsequent Steps:** After compaction, the fabric may undergo calendaring to enhance its hand-feel, making it "hot and soft". It then moves into the final quality control phase, which includes laboratory testing and a detailed inspection using a 4-point review system.
- **Calendaring**
This process takes place after the fabric has passed through the compactor. The primary objective of calendaring is to **make the fabric "hot and soft,"** enhancing its final hand-feel and aesthetic appeal.

Ultimately, if the fabric passes these final reviews after compaction, it is approved for delivery; otherwise, it is sent for reprocessing.

CHAPTER 5

Fabric Inspection

5.1 Introduction

Fabric inspection is a critical quality control process carried out to ensure that the produced fabric meets the required standards before it is sent for further processing or delivery. In this stage, the fabric is carefully examined to identify defects such as holes, stains, uneven dyeing, knitting faults, and other irregularities that may affect its quality and performance.

The inspection is usually performed using an inspection machine where the fabric is passed over a well-lit surface, allowing inspectors to visually detect defects. Standard systems such as the 4-point inspection system are often used to evaluate and grade the fabric based on the severity of defects. During this process, parameters like fabric width, GSM (grams per square meter), shade consistency, and overall appearance are also checked.

Fabric inspection helps in maintaining consistent quality, reducing customer complaints, and ensuring that only acceptable fabric proceeds to the next stage. It plays a vital role in improving production efficiency and upholding the reputation of the manufacturer.

5.2 Table Inspection

Fabric inspection is first carried out on an inspection table under adequate lighting conditions to ensure clear visibility of the fabric surface. The fabric is spread on the table and checked manually by inspectors for various quality parameters. This initial inspection focuses on identifying issues such as running shade, shade variation, stains, holes, oil marks, knitting faults, and other visible defects.

During this process, inspectors carefully observe the fabric for color consistency across its length and width, ensuring there is no variation in shade. The fabric surface is also examined for physical defects and irregularities that may affect its appearance or performance. This method allows quick detection of major defects before the fabric is passed to machine inspection.

Table inspection plays an important role in maintaining quality control by ensuring that defective fabric is identified early in the process, reducing further processing losses and ensuring better overall product quality.

5.3 4-Point System Inspection

The **4-point system** is a standardized method used during fabric inspection to quantify quality by assigning penalty points to defects based on their size.

The 4-Point Scoring System

Points are assigned to individual defects according to their length or nature:

- **1 Point:** Defects up to 3 inches.
- **2 Points:** Defects longer than 3 inches but up to 6 inches.
- **3 Points:** Defects longer than 6 inches but up to 9 inches.
- **4 Points:** Defects longer than 9 inches.
- **4 Points (Special):** Any **hole**, regardless of size, is automatically assigned the maximum of 4 points.

Defects Monitored

The inspection sheet tracks a wide variety of specific faults, including:

- **Physical/Structural:** Thick & Thin, Slubs, Miss Stitch, Needle Striker, and No. of Cut.
- **Surface/Handling:** Crease Marks, Crumples, Edge Marks, and Dirt Marks.
- **Dyeing/Chemical:** Patchy/Uneven color, Dye Stains, and Softener Spots.
- **Contamination:** Oil Spots and "Dead Cotton".

5.4 Rejection Criteria

The final grade of the fabric is determined by the **Point Rate**, calculated using the formula:

(Total Points x 3,600) / (Roll Length in Yards x Actual Width in Inches).

Based on the acceptance calculation:

Grade A (Pass): Up to 20 points.

Grade B (Pass): 21 to 28 points.

REJECT: Any fabric scoring **above 28 points** is rejected

Points per 100 yd² Formula

$$\text{Points per 100 yd}^2 = \frac{\text{Total points in roll} \times 36 \times 100}{\text{Fabric length (yards)} \times \text{Fabric width (inches)}}$$

Additionally, a roll may be rejected regardless of the point score if it fails specific Remarks checks, such as **Hand Feel**, **Running Shade**, or if it is flagged as **Not OK** for AQL (Acceptable Quality Level) and ITP (Inspection Test Plan) standards.

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FABRIC INSPECTION REPORT (4 POINT SYSTEM)																																																																	
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Figure 13 4 Point Inspection Sheet

CHAPTER 6

Yarn Dyeing

6.1 Introduction

The yarn dyeing facility operates with a total plant capacity of **15,000 kg per day**, split between 12,500 kg of Cotton and 2,500 kg of Acrylic. This production is supported by 32 dyeing machines and a robust utility infrastructure, including:

- **Boilers:** 3 units with a capacity of 6 ton/hr (BOSCH).
- **Generators:** 4 units (1 Kohler at 1640KW and 3 Perkins/Cummins at 800KW).
- **Water Management:** A Water Treatment Plant (WTP) providing 6,000 m³/day and an Effluent Treatment Plant (ETP) processing 3,500 m³/day.



Figure 14 Yarn Dyeing

6.2 Yarn Dyeing Flow Chart

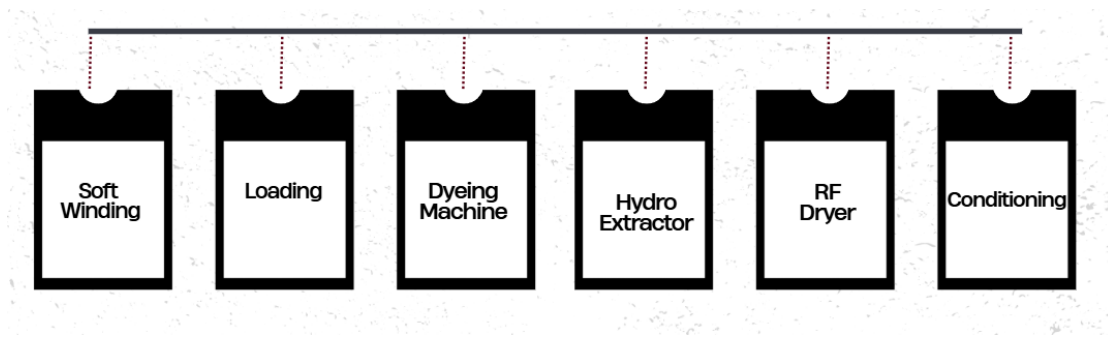


Figure 15 Yarn Dyeing Flow Chart

The workflow follows a precise sequence to ensure uniform color and fiber integrity:

Preparation and Soft Winding

The process begins with Raw Yarn In-house arrival, followed by a QC & Lot Check to ensure material consistency before entering the Dyeing Plan & Machine Plan phase,

Soft Winding: Using SSM machines (6 units, 492 spindles) operating at 800 RPM, yarn is rewound into perforated dye tubes. This step is critical because it creates a soft package with specific density, allowing dye liquor to circulate evenly through the yarn layers



Figure 16 Soft Winding Machine

Pre-Treatment

Before dyeing, the yarn undergoes Batch Preparation and is loaded into the dyeing vessels,

- **Scouring and Bleaching:** These chemical processes remove natural impurities, waxes, and pectin from fibers like cotton.
- **Pre-wetting and Rinsing:** The yarn is prepared to absorb chemicals uniformly by ensuring the fibers are fully saturated

The Dyeing Cycle:

Utilizing machines from Thies (14 units), Tong Geng (6 units), and Gofront (6 units), the dyeing phase involves:

- **Circulation:** A continuous In-Out pump circulation cycle (typically 6 min/4 min).
- **Temperature Control:** Controlled heating and pressure rise to facilitate dye bonding.
- **Dosing:** Precise injection of dyes, salts, and alkalis. The facility uses Reactive dyes for Cotton and Viscose, Acid dyes for Nylon, and Basic dyes for Acrylic.



Figure 17 Yarn Dyeing Machine

After-Treatment:

- Hydro Extraction: Using two DETTIN machines, high-speed centrifugal force spins the yarn packages to expel free moisture in an 8-minute cycle.



Figure 18 Hydro Extractor Machine

- RF Drying: Three STALAM Radio Frequency dryers heat moisture volumetrically from the core of the yarn to the surface. This is faster and more uniform than hot-air drying, taking 45 minutes to 1.5 hours.



Figure 19 RF Dryer Machine

6.3 Final QC and Finishing

- **Quality Control (QC):** A final check is performed to verify color matching and lot consistency.
- **Hard Winding:** The dyed yarn is rewound from the soft dye packages onto final "hard" paper cones suitable for knitting or weaving.
- **Packing:** The finished product is then packaged for delivery to the client.

CHAPTER 7

Washing Section

7.1 Introduction

The washing process in SQ Hues production is a critical finishing stage used to achieve specific textures, clean the fabric, and enhance its surface quality.

7.2 Types of Washing Processes

The facility utilizes several specialized washing techniques:

- **Normal Wash:** This process primarily focuses on fabric texture. It uses **cationic and silicon softeners** to provide a **"slippery" or "warm soft" finishing** to the material.
- **Acid Wash:** Typically performed **after dyeing**, this wash is used for 100% cotton and 100% acrylic (to achieve a "cashmery-like" feel). It involves the use of **reactive dischargeable dyes** and **potassium permanganate**.
- **Enzyme Wash:** This is used for 100% cotton and "schoolware" to remove **extra fibers** and improve the **surface finish**.

The technical process includes:

- Adjusting the bath to **pH 4.5** using chemicals and acid.
- **Enzyme dosing** at 50–55°C.
- An **"enzyme kill"** step at 80°C, followed by a **cold wash**.
- This process typically uses a **liquor ratio of 1:18**.

7.3 Washing Flow Chart

Sample & Bulk Approval: Initial samples are tested and approved to ensure the desired color and quality before starting bulk production.

Chemical Arrangement: Required dyes and chemicals are prepared and measured according to the recipe for accurate dyeing results.

Machine Loading: Fabric is loaded into the dyeing machine properly to ensure smooth processing and uniform treatment.

Washing Process: The fabric is washed to remove excess dyes and chemicals, improving cleanliness and color fastness.

Hydro Extraction: Excess water is removed from the fabric using a hydro extractor to reduce moisture content before drying.

Drying: The fabric is dried under controlled conditions to achieve the required moisture level and stability.

Inspection: Final fabric is checked for defects, shade consistency, and overall quality before approval for further use or delivery.



Figure 20 Washing Flow Chart

7.4 Washing Machinery and Capacity

The washing department uses several high-capacity and specialized machines:

YILMAK: Used for both bulk (up to **400kg**) and sample production. The facility's average output is noted at over **21 tons**.

Tonello: A machine with a **200kg capacity** that is specifically highlighted for **cost-saving** because it requires less water and fewer chemicals.

Low Capacity YILMAK: Smaller machines used for sampling, with capacities ranging from **5kg to 50kg**.



Figure 21 Washing Machine

7.5 Post-Wash Finishing and Drying

After washing, the material must be dried at temperatures specific to the fiber type to prevent damage or shrinkage:

- **Cotton:** 90–100°C.
- **Viscous:** 70–80°C.
- **Acrylic:** 60–70°C.

For larger volumes, the facility uses specialized dryers like the **TRIVENETA** (80–100kg capacity) and various **YILMAK Dryers** (ranging from 10kg up to 120kg)



Figure 22 Drying Machine

CHAPTER 8

Color Development Lab

8.1 Introduction

The color development laboratory in the textile industry is a specialized unit responsible for creating and matching fabric colors according to customer requirements. It plays a vital role in ensuring accurate shade development before bulk production begins. In this lab, small fabric samples are dyed using controlled conditions to develop the desired color through precise combinations of dyes and chemicals.

The lab uses advanced equipment and standardized procedures to maintain consistency, reproducibility, and quality of shades. Parameters such as temperature, time, pH, and chemical concentration are carefully controlled during the process. Once the required shade is achieved and approved, the recipe is transferred to bulk production. Thus, the color development lab ensures shade accuracy, reduces production errors, and helps maintain overall product quality.

8.2 Color Development Lab Workflow

The laboratory operates through a systematic, data-driven workflow:

1. Digitization and Recipe Formulation

- **Analysis:** The process begins with a physical swatch, Pantone reference, or QTX file, which is analyzed for composition and texture.
- **Scanning:** A **Spectrophotometer** scans the swatch to translate the color into digital data.
- **Software Processing:** Specialized software (Taps) evaluates this data based on **dye costs, metamerism, and strength** to formulate the optimal recipe.

2. Automated Robotic Dispensing

- Once a recipe is selected, digital instructions are sent to an **Automated Color Dispensing Robot**.
- The robot is equipped with **128 individual color tubes** and uses precision pipettes to dispense the exact amounts of dyes, salts, soda, caustic, and auxiliaries required, removing the risk of manual weighing errors.

3. Lab-Dip Dyeing and Finishing

- The dispensed ingredients are combined with the substrate (such as fabric, yarn, acrylic, or nylon) and loaded into a dyeing machine.
- **Controlled Drying:** After dyeing and washing, the material undergoes padding (squeezing) and drying at specific temperatures: **90–100°C for cotton** or **60–70°C for acrylic**.

4. Verification and Bulk Handover

- **Validation:** The resulting **lab-dip sample** is verified against the original standard. If the shade deviates, the automated cycle repeats until an exact match is achieved.
- **Production Recheck:** Before large-scale production begins, the production team provides the actual yarn lot for a final double-check against the approved hanger to ensure consistency during bulk manufacturing



Figure 23 Color Lab

CHAPTER 9

Project at SQ Hues Ltd

9.1 Introduction

This report presents a comprehensive Root Cause Analysis (RCA) of fabric rejection occurring in the Circular Knitting Section. The analysis covers a four-month period spanning December, January, February, and March, during which a total of 1,286.8 kg of fabric was rejected - representing significant production loss in terms of material, labor, time, and cost.

The primary objective of this study is to identify the dominant causes of fabric rejection, determine the root cause through systematic quality tools, and propose actionable corrective and preventive measures to minimize future wastage.

9.2 Key Findings at a Glance

Total Rejection	1,286.8 kg over December – March (4 months)
Top Defect	Spandex Cotton Mixing — 472.2 kg (36.7% of total rejection)
Root Cause	Lack of standardized machine maintenance and feeder adjustment procedures
Analysis Tools Used	Pareto Chart, Pie Chart, Fishbone Diagram, 5-Why Analysis, Kaizen Model
Forecast	37% → 25% rejection rate in 6 months; further down to 20% in 12 months

Table 1 Key Findings

The Spandex Cotton Mixing defect alone accounts for more than one-third of total rejection, making it the single most critical quality issue in the knitting section. The 5-Why analysis traced this defect back to a lack of standardized preventive maintenance scheduling and feeder adjustment protocols. By implementing the five proposed corrective actions, the facility is projected to reduce its rejection rate from 37% to 20% within twelve months of implementation.

9.3 Overview of Production Rejection

During the four-month period from December to March, the circular knitting section generated 1,286.8 kg of rejected fabric material. This figure represents a substantial portion of total production output, indicating that the section is operating well below optimal quality standards. Rejected fabric results in direct material loss, additional labor costs for inspection and handling, machine downtime, and delayed order fulfillment.

The rejection data was collected from the fabric inspection process downstream of the knitting machines. Each roll of fabric was inspected upon completion, and defects were recorded, categorized by type, and measured in kilograms. The aggregated data was then analyzed using pivoted tables and bar charts to identify patterns and priorities.

9.4 Defect Breakdown

The table below presents the complete breakdown of all identified defect categories, their respective volumes, percentage contributions, and priority ranking. This data forms the foundation of the Pareto analysis conducted in Section 4.

Defect Type	Rejected Qty (kg)	% of Total	Priority Rank
Spandex Cotton Mixing	472.2	36.7%	#1 — Critical
Needle Broken	224.3	17.4%	#2 — High
Spandex Drop	200.3	13.7%	#3 — High
Lycra Missing	135.9	10.6%	#4 — Medium
Spandex Filamentation	74.8	5.8%	#5 — Medium
Broken Needle	54.8	4.3%	#6 — Low
Spandex Missing	57.2	4.4%	#7 — Low
Vertical Tuck Stitch	~20	~1.6%	#8 — Low
Wheel Free / Line Star / Others	~47.3	~3.5%	#9+ — Minor

Table 2 Defect Breakdown

9.5 Key Observations

- The top four defect types are Spandex Cotton Mixing, Needle Broken, Spandex Drop, and Lycra Missing together account for approximately 78.4% of total rejection, consistent with the Pareto principle.

- Spandex-related defects (Spandex Cotton Mixing, Spandex Drop, Spandex Filamentation, Spandex Missing) collectively account for over 58% of total rejection, pointing strongly to a systemic issue with spandex/Lycra handling and feeding.
- Needle-related defects (Needle Broken, Broken Needle) together account for approximately 21.7% of total rejection, suggesting inadequate needle maintenance and inspection routines.
- The majority of defects are machine related or process-related, indicating that improvements in machine maintenance, operator training, and process standardization will have the greatest impact on rejection reduction.

9.6 Pareto Principle in Quality Management

The Pareto Principle, commonly known as the 80/20 rule states that approximately 80% of problems are caused by 20% of the causes. In quality management, Pareto analysis is used to prioritize corrective actions by identifying the **vital few** causes that drive the **trivial many** instances of defects. By focusing resources on the most impactful causes, organizations can achieve the greatest improvement with the least effort.

9.7 Pareto Chart Analysis

The Pareto chart constructed from the defect data (Total Reject Qty = 1,286.8 kg) shows the following cumulative pattern:

Defect	Qty (kg)	% Share	Cumulative %	Zone
Lycra Cotton Mixing	472.2	36.7%	36.7%	A Zone
Needle Broken	224.3	17.4%	54.1%	A Zone
Spandex Drop	200.3	13.7%	67.8%	A Zone
Lycra Missing	135.9	10.6%	78.4%	A Zone
Spandex Filamentation	74.8	5.8%	84.2%	B Zone
Broken Needle	54.8	4.3%	88.5%	B Zone
Spandex Missing	57.2	4.4%	92.9%	B Zone
Others	~67.3	~7.1%	100%	C Zone

Table 3 Pareto Analysis

9.8 Pareto Interpretation and Focus Area

The Pareto analysis clearly identifies the first four defect types as the **A Zone** the critical zone requiring immediate corrective action. These four defects together account for 78.4% of all

rejection, confirming that addressing these root causes will produce the highest improvement impact.

Among these, Spandex Cotton Mixing stands out as the single largest contributor at 36.7%, making it the primary focus of this root cause analysis. The decision to conduct detailed Fishbone and 5-Why analysis on this defect is well-supported by the Pareto data.

It is also noteworthy that Spandex Drop (13.7%) and Lycra Missing (10.6%) are closely related to the same underlying cause improper Spandex/Lycra feeding and handling suggesting that addressing the root cause of Spandex Cotton Mixing will likely produce improvement across multiple defect categories simultaneously.

9.9 Definition of the Defect

Spandex Cotton Mixing defect is a fabric quality issue that occurs when cotton yarn and spandex (Lycra/Elastane) yarn are not properly combined during the circular knitting process, resulting in areas where spandex is missing or unevenly distributed due to feeder tension issues, misalignment, yarn drop, or machine malfunction. This defect is identified by uneven stretch, distorted loops, visible streaks, fabric puckering, and inconsistent hand-feel across the fabric. Such defective fabric fails to meet quality standards and is either downgraded or rejected, leading to significant production loss. Over a four-month period, this defect alone caused 472.2 kg of fabric rejection, resulting in substantial financial loss and wasted processing potential, as the fabric cannot proceed to subsequent stages like dyeing and finishing.

9.10 Root Cause Analysis

The Fishbone Diagram also known as the Ishikawa Diagram or Cause-and-Effect Diagram is a structured quality analysis tool used to identify all possible causes of a specific quality problem. Developed by Professor Kaoru Ishikawa, this method organizes potential causes into major categories, helping teams move beyond symptoms and surface-level observations to understand underlying drivers of defects.

For this analysis, the five standard categories applied are: Man (human factors), Machine (equipment-related), Material (raw material quality), Method (process and procedures), and Environment (working conditions). Each category was systematically examined for contributing causes to the Spandex Cotton Mixing defect.

9.11 Cause Analysis by Category

Category	Contributing Causes
Man (Human Factor)	• Lack of operator skill in handling Lycra/spandex thread • Incorrect yarn tension setting by operators • Improper Lycra thread handling techniques • Poor machine monitoring during active production • Negligence in checking needle condition before and during production
Machine	• Needle damage or wear leading to irregular knitting • Sinker damage causing improper fabric loop formation • Lycra feeder malfunction resulting in inconsistent feeding • Needle bed damage affecting overall fabric uniformity • Lycra drop due to improper feeder and tension settings
Material	• Different yarn lot mixing causing uneven stretch • Cotton yarn unevenness in yarn count or twist • Low quality Lycra yarn with poor elastic recovery • Lycra missing or improperly loaded on the machine • Uneven stretch and irregular elastic properties in yarn
Method	• Incorrect machine speed settings for specific fabric constructions • Poor yarn feeding method causing inconsistent tension • Uneven Fabric GSM due to improper machine settings • Improper needle replacement schedule and practices
Environment	• Dust and lint accumulation in the knitting area • Poor lighting conditions for quality inspection • Static electricity buildup in yarn causing misfeeding • Unstable temperature and humidity affecting yarn behavior

9.12 Most Critical Causes Identified

Following the multi-category brainstorming, the fishbone analysis team identified the following as the most probable and high impact causes for the Spandex Cotton Mixing defect:

- **Lycra feeder malfunction / misalignment:** The most direct mechanical cause. When the feeder does not correctly control the spandex yarn path, drop or mixing defects occur instantly.

- **Incorrect Lycra tension setting:** If the tension is too high, the spandex yarn breaks or skips feeders. If too low, the yarn feeds inconsistently, causing uneven integration with cotton.
- **Lack of operator skill in Lycra thread handling:** Improper loading or re-threading of the spandex yarn can create incorrect feeding paths that lead to defects mid-production.
- **Low quality Lycra yarn / mixing different yarn lots:** Variations in yarn properties between lots cause unpredictable behavior in the knitting machine, making consistent quality impossible.

9.13 Root Cause from Fishbone Analysis

ROOT CAUSE IDENTIFIED FROM FISHBONE:	
Incorrect Spandex Feeding due to improper feeder adjustment in the knitting machine, which leads to unstable yarn feeding and results in Lycra drop.	

9.14 5 Why Analysis

The 5-Why technique is a systematic approach to root cause analysis that involves asking **Why?** repeatedly typically five times to trace a problem from its observed symptom all the way back to its fundamental root cause. Developed as part of the Toyota Production System, this technique is widely applied in lean manufacturing and quality management to prevent recurring problems by addressing the root cause rather than just the symptoms.

The principle underlying 5-Why is that the first few answers to **Why?** often reveal symptoms or secondary causes, while the deeper answers typically the 4th or 5th Why reveal process failures, system gaps, or organizational deficiencies that, when corrected, permanently eliminate the problem.

9.15 Why Chain for Spandex Cotton Mixing Defect

Step	Question Asked	Answer / Finding
Why 1	What is the visible symptom of the problem?	Lycra yarn and cotton yarn are not entering the knitting machine smoothly together, causing mixing defects in the finished fabric.

Why 2	Why are they not entering smoothly?	The Lycra yarn tension becomes unstable during knitting, causing the yarn to slip, skip, or drop at various feeder points on the circular knitting machine.
Why 3	Why does the tension become unstable?	The Lycra yarn feeder and tension controller in the knitting machine are not set correctly for the specific yarn count and machine speed in use.
Why 4	Why are they not set correctly?	The machines are not checked regularly for feeder alignment and tension calibration. There is no structured routine for verifying machine settings before production begins.
Why 5 (Root Cause)	Why are machines not checked regularly?	There is no standardized preventive maintenance schedule or documented operating procedure for feeder adjustment and tension verification.

9.16 Root Cause Summary

CONFIRMED ROOT CAUSE: Lack of a Standardized Preventive Maintenance Schedule and Feeder Adjustment Protocol The absence of a structured, documented maintenance routine means that machine feeders, tension controllers, and alignment settings are not regularly verified or corrected. This allows gradual drift in machine settings to go undetected until a defect appears in the fabric — at which point significant material has already been wasted.	
--	--

The 5-Why analysis reinforces and confirms the Fishbone analysis findings. Both tools independently point to the same systemic deficiency: the absence of standardized, documented procedures for machine maintenance and feeder calibration. This is a process and management gap, not a one-time machine failure which means the solution must also be systemic (a new process or system), not a one-time failure.

9.17 Proposed Solution

Based on the root cause analysis conducted through Fishbone and 5-Why methodologies, the following five solutions are proposed to address the Spandex Cotton Mixing defect and other related rejection causes. These solutions target the root cause directly (Lack of standardized maintenance and feeding procedures) as well as contributing causes across Man, Machine, Material, Method, and Environment categories.

9.18 Solution 1: Ensure Proper Spandex Feeding

Category	Man / Method
----------	--------------

- Train all machine operators on the correct Spandex/Lycra feeding path specific to each machine type.
- Ensure yarn passes correctly through all guides, tensioners, and feeders in the prescribed sequence.
- Operators must verify yarn alignment and tension before starting every new production batch.
- Create a visual SOP (Standard Operating Procedure) chart and display it near each knitting machine.
- Supervisors should periodically observe and verify correct feeding technique during production.

9.19 Solution 2: Monitor and Prevent Dust in the Knitting Area

Category	Environment
----------	-------------

- Implement a daily housekeeping schedule to remove dust and lint from the production floor.
- Assign dedicated workers to inspect and clean yarn guides, feeders, and tension controllers each shift.
- Install air filtration or suction systems near knitting machines to reduce airborne lint.
- Conduct a weekly deep cleaning of all machines to prevent accumulation in mechanical parts.
- Document all cleaning activities in a maintenance logbook for traceability.

9.20 Solution 3: Perform Quality Checking at Specific Time Intervals

Category	Method / Man
----------	--------------

- Inspect knitted fabric at standardized time intervals (every 30 minutes during production).
- Check specifically for Lycra drop, uneven stretch, loop defects, and fabric GSM variations.

- Record all inspection results in a quality monitoring sheet with timestamp, machine ID, and operator name.
- Any defect detected during in-process inspection should trigger immediate machine stoppage.
- Use fabric swatch samples to compare output quality against the approved standard sample.

9.21 Solution 4: Ensure Good Quality Spandex Yarn

Category	Material
----------	----------

- Purchase Spandex/Lycra yarn only from pre-qualified, reliable suppliers with consistent quality history.
- Conduct incoming yarn quality inspection before releasing any new yarn lot for production use.
- Check yarn denier, elongation, elastic recovery, and uniformity as part of incoming QC.
- Avoid mixing yarn from different lots on the same machine or within the same production run.

9.22 Solution 5: Worker Reward System (Monthly & Annually)

Category	Man / HR
----------	----------

- Recognize and reward the top 3 workers monthly based on: maximum production output, minimum fabric waste, and minimum needle breakage incidents.
- Rewards include a cash bonus from the HR Department and public recognition (photo displayed on the production floor dashboard).
- Annual salary increases to be tied directly to the monthly performance score of each worker.
- This initiative creates healthy competition and intrinsic motivation among workers to follow best practices.
- Track individual performance with a transparent KPI board visible to all production staff.

9.23 KAIZEN Improvement Model

Kaizen a Japanese term meaning **continuous improvement** is a philosophy and practical methodology used in manufacturing to achieve sustained quality and efficiency gains through small, incremental improvements applied consistently over time. Unlike large scale re-engineering projects, Kaizen focuses on building a culture of problem-solving, where every worker at every level actively identifies and eliminates waste and defects.

The Kaizen Improvement Model applied in this report follows the structured cycle of:

Identifying the problem → analyzing the cause → defining actions → implementing changes → measuring results → standardizing improvements.

This model ensures that the improvements made in response to the Spandex Cotton Mixing defect are not temporary fixes but become embedded in the facility's standard operating procedures.

9.24 Kaizen Model Application

PROBLEM	Spandex Cotton Mixing Defect The highest contributor to production waste at 36.7% of total rejection (472.2 kg out of 1,286.8 kg) over four months.
ROOT CAUSE	Incorrect Spandex Feeding due to improper feeder adjustment in the circular knitting machine, leading to unstable Lycra yarn tension and resulting in Lycra drop defects.
KAIZEN ACTIONS	1. Train workers on correct Lycra/Spandex feeding path 2. Regular machine cleaning and dust control 3. Follow standardized yarn feeding procedure 4. Conduct periodic quality checks every 30 minutes 5. Use only high-quality Spandex yarn from certified suppliers
EXPECTED RESULTS	Stable Lycra feeding • Reduced Lycra drop defects • Improved fabric elasticity and consistency • Better overall knitting machine performance • Fewer defect-related stoppages
BENEFITS	Reduced fabric rejection rate from 37% to 20% • Higher production efficiency • Lower raw material wastage • Cost savings on labor, raw material, and rework • Continuous improvement culture in the knitting section

9.25 Implementation Notes

For the Kaizen model to be effective, senior management commitment is essential. The proposed actions — particularly operator training, maintenance scheduling, and the worker reward system

require top-down support and resource allocation. A designated Kaizen champion or quality team should be assigned to monitor implementation progress, track key performance indicators (KPIs), and ensure continuous adherence to the new standards.

Key performance indicators to monitor during implementation should include daily rejection rate per machine, number of Spandex Cotton Mixing defects per shift, operator training completion rates, maintenance schedule adherence rate, and fabric GSM consistency metrics.

9.26 Reject Reduction Forecast

The reject reduction forecast is based on the projected impact of the five proposed solutions on the primary defect (Spandex Cotton Mixing at 36.7%) and related defects. The forecast assumes phased implementation: solutions 1–3 (operator training, dust control, quality checking) will be implemented in the first 1–3 months, with solutions 4–5 (yarn quality control and reward system) fully operational by month 6. The full benefits of all solutions combined are expected to materialize by the 12-month mark.

9.27 Rejection Rate Forecast Table

Timeline	Est. Rejection Volume	Rejection Rate	Material Saved	Status / Notes
Current (Baseline)	1,286.8 kg / 4 months	37%	—	Active — No corrective action applied
6 Months Post-Implementation	~811.7 kg projected	25%	~475.1 kg saved	Solutions partially implemented and monitored
12 Months Post-Implementation	~643.4 kg projected	20%	~643.4 kg saved	Full solution deployment with sustained improvement

9.28 Material Savings Projection

If the facility successfully reduces its rejection rate from the baseline of 37% to the 12-month target of 20%, the material savings are estimated as follows:

- Current average monthly rejection: approximately 321.7 kg/month (1,286.8 kg ÷ 4 months).
- At 20% rejection rate, monthly rejection would reduce to approximately 173.7 kg/month (a reduction of ~148 kg/month).
- Annual material savings: approximately 1,776 kg/year of fabric material recovered.
- In addition to material savings, labor and machine time savings from fewer defect-related stoppages and rework activities will further improve overall production efficiency.

9.29 Conclusion

This Root Cause Analysis has systematically investigated the fabric rejection problem in the Circular Knitting Section, analyzing 1,286.8 kg of rejected material over four months. Through the structured application of Pareto Analysis, Fishbone Diagram, and 5-Why methodology, the investigation has arrived at a clear, actionable root cause.

The Pareto analysis confirmed that Spandex Cotton Mixing is the dominant defect, contributing 36.7% of all rejections. The Fishbone analysis identified incorrect spandex feeding due to improper feeder adjustment as the primary mechanical cause. The 5-Why analysis then traced this back to the fundamental process gap: the absence of a standardized preventive maintenance schedule and feeder adjustment protocol.

This finding is significant because it shifts the solution from a reactive, machine-fixing approach to a proactive, system-building approach. The problem is not that the machines broke down randomly, it is that there was no system in place to routinely verify and maintain optimal machine settings. Building that system is the core of the solution.

9.30 Recommendations

1. Immediate Actions (Within 1 Month)

- Conduct a mandatory operator training session on correct Spandex/Lycra feeding techniques for all knitting machine operators.

- Develop and display visual SOPs (Standard Operating Procedures) for Spandex feeding at each machine station.
- Initiate daily housekeeping routines for the knitting area, focusing on dust and lint removal from yarn feeders and guides.

2. Short-Term Actions (1–3 Months)

- Implement a structured in-process quality inspection system with checks every 30 minutes per machine.
- Develop and implement a Preventive Maintenance (PM) schedule for all circular knitting machines, with specific focus on Lycra feeder alignment and tension calibration.
- Review and standardize Spandex yarn supplier qualification criteria and incoming quality inspection procedures.

3. Medium-Term Actions (3–6 Months)

- Launch the Worker Reward System with clearly communicated criteria (maximum production, minimum wastage, minimum needle break) and recognition events.
- Review the KPI dashboard monthly to track rejection rate trends per machine and per defect type.
- Conduct a review of yarn lot management to implement strict lot segregation procedures in the warehouse.

4. Long-Term / Continuous Improvement

- Institutionalize the Kaizen philosophy in the knitting section through regular quality circles and improvement workshops.
- Target a 20% rejection rate by month 12, then continue to drive further improvements beyond that benchmark.
- Expand the root cause analysis methodology to other sections of the facility (dyeing, finishing) to replicate the improvement culture.

CHAPTER 10

Conclusion

10.1 Conclusion

This industrial attachment at SQ Hues Limited provided comprehensive practical exposure to modern textile manufacturing processes, particularly in knitting, dyeing, finishing, washing, and quality control operations. The study highlighted how vertical integration and advanced machinery contribute to maintaining high product quality and operational efficiency across the production chain.

A detailed analysis of the circular knitting section revealed that fabric rejection is a significant issue, with a total of 1,286.8 kg rejected over a four-month period. Among all defects, Spandex Cotton Mixing was identified as the most critical, contributing 36.7% of total rejection. Through systematic application of quality tools such as Pareto analysis, Fishbone diagram, and 5-Why analysis, the root cause was traced to the lack of standardized preventive maintenance and improper feeder adjustment procedures.

The findings emphasize that most defects are machine- and process-related, indicating that improvements in maintenance practices, operator training, and process standardization can significantly reduce rejection rates. Implementation of the proposed corrective measures is expected to enhance production efficiency, minimize material waste, and improve overall product quality.

In conclusion, this study not only provided valuable industrial experience but also demonstrated the importance of systematic problem-solving approaches in identifying root causes and improving manufacturing performance in the textile industry.